# P1-09 — Gate G1 Novelty Closure Audit

**Command:** P1-09 — Gate G1 Novelty Closure Audit **Date:** 21 October 2026 **Companion:** P1_09_OVERLAP_MATRIX.md (claim-by-claim matrix K1–K7 × L1/L2/L3/L4/L8/neighbours, with per-source records).

**Gate G1 recommendation: DEFER — Gate G1 cannot be closed on the present evidence.** Overall project decision: **CONDITIONAL GO**. Recommendation on novelty claims: **NO-GO for any priority/"first" claim** until repo L1 is read in full. This audit is not permission to draft the manuscript.

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## 1. Freeze verification (Step 1)

- Branch arena/ab8e330d-galios-hull; the local checkout was found reset to base commit 33aeb79 with the project files untracked (sandbox state, again), while the remote held the full pushed history. The workspace was first backed up (/tmp/proj_backup_p109), the remote state restored by fast-forward to origin/arena/ab8e330d-galios-hull, and diff -r confirmed the restored tree is **identical** to the pre-reset workspace (no loss, no history rewrite, no force-push).
- Current commit (before this audit's commits): 33a8c28 — "P1-08: integration audit". Working tree clean at verification time.
- Frozen blueprint 02_THEOREM_BLUEPRINT.md SHA-256: ec2fd8d58f8d2a986314f83997d750d689f7c88e98ec8f87753ab4c9cf438cff — identical to the after-hash recorded in P1_08_INTEGRATION_AUDIT.md. **The blueprint was not edited in this command**, and no proof or validation artifact was modified.

## 2. The L1 question (Step 2) — still unresolved, lawfully

**Exact paper.** Indibar Debnath, Om Prakash & Habibul Islam, "Galois hulls of constacyclic codes over finite fields", ∗Cryptography and Communications∗ 15(1), 111–127 (2023). DOI 10.1007/s12095-022-00591-6 (published online 23 June 2022; issue Jan 2023). Correction: DOI 10.1007/s12095-022-00602-6, 15, 129–130 (2023). zbMATH Zbl 1518.94136 (paper), Zbl 1518.94137 (correction). OpenAlex W4283360609; Semantic Scholar CorpusId 250008629; DBLP journals/ccds/DebnathPI23. Authors at IIT Patna.

**Version history (verified):** journal version of record (Springer, 2022/2023) + a 10-item correction (2023, lawfully read in full during P1-03 — all items editorial as displayed; the displayed B_i factor-set definitions mirror orbit-style bookkeeping but do not settle scope). No preprint (arXiv or other repository) has been located in any search to date.

**Lawful-access attempts (fresh, 21 Oct 2026, plus the P1-02/P1-03 records):**

- OpenAlex API record for the DOI (fresh fetch): open_access.is_oa = false, oa_status = closed, any_repository_has_fulltext = false, no PDF URL.
- Semantic Scholar Graph API (fresh fetch): isOpenAccess = false, openAccessPdf.url empty.
- Springer article page (lawful): abstract + metadata only; full text paywalled. A fresh web search (query: the exact title + preprint/manuscript/pdf, excluding ResearchGate) returned only the Springer page, the correction, an academia.edu copy **of the correction only** (unverified licence — not used), and citing papers. No author manuscript, institutional copy, or repository version was found.
- Not attempted (correctly): paywall circumvention of any kind; inferring theorems from the abstract.

**What the abstract level does verify:** "This paper presents a formula for the dimension of Galois hulls of constacyclic codes" (Springer abstract), and — corroborated by the introduction of repo L4 (ScienceDirect snippet, 21 Oct 2026) — "they also counted the number of constacyclic codes of a given Galois hull dimension."

**Exactly which comparisons therefore cannot yet be made (the L1-Q list):** (Q1) Does L1 allow

p | n (repeated-root lengths) or assume gcd(n, p) = 1? (Q2) Is its hull-dimension formula equivalent to our finite-field specialization of T4 (min(P − α_i, α_{i−1}) edge form), and under which hypotheses on q, k, λ? (Q3) Is its count a count of **labeled codes by hull dimension only**, or joint in (dim C, dim Hull)? (Q4) Does it count codes at all in the affine/product setting (title says finite fields — presumably not, unverified)? (Q5) Does it contain means/variances? (Q6) How do its B_i sets relate to τ-orbits (relevant to our C3/C4 counts)? Until a lawful full text is read, **no overlap verdict stronger than "unresolved" is honest for K1, K2, K4 against L1**.

## 3. Overlap assessment (Step 3) — summary

Full details: P1_09_OVERLAP_MATRIX.md. Headlines, with evidence levels:

- **L4 (affine; arXiv full text verified):** substantial partial overlap with **K1** — hull generator (Thm. 7), dual generator (Thm. 5), and a repeated-root hull-dimension theorem (Thm. 8, arXiv numbering) whose b-terms are cyclic sums of max{u_δ, p^{e′} − u_{δ−1}} expressions, **under extra hypotheses** (gcd(ord λ, n′) = 1, ord λ | 1 + p^{e−k}) and in an idempotent (semisimple) component presentation. It counts attainable dimension ∗values∗ (Cor. 3.5), not codes, and its conclusion leaves code enumeration over A open. No joint enumerator, transfer trace, or moments in the inspected text.
- **L8 (multi-twisted/GPM; arXiv full text verified):** strong hull-machinery overlap in a different family (Thms. 8–11, 21–22, 25–29, arXiv numbering); constacyclic codes are the index-1 case with no general simple-root restriction found — so T3/T4-style hull formulas are **not** first-in-field. No joint enumerator/transfer/moments in the inspected text.
- **L2 (average dimensions; abstract verified):** owns the ∗mean-as-headline∗ territory over finite fields/R_{m,q}. Our C5 must stay a corollary (GAP-12), never a headline.
- **L3 (small hull dimensions; publisher page verified):** covers hull dimensions 1–2 including p | n cases — "first repeated-root hulls" would be false.
- **Neighbours (scope level):** repeated-root hull/LCD literature and classical hull enumeration/averages exist; the inspected Check-D records (P1-02 §3, plus a fresh targeted search on 21 Oct 2026 returning no counterpart) contain **no** labeled affine repeated-root joint (dim C, dim Hull) transfer-trace enumerator. Databases: arXiv, zbMATH Open, OpenAlex, Semantic Scholar, publisher pages. **Not searched: MathSciNet, Scopus, Web of Science** (no lawful access). Non-exhaustive; non-discovery ≠ novelty.

**Net assessment.** The mathematical centre that no ∗inspected∗ source contains is the package **K2 + K3 + K6** (joint labeled enumerator over the affine product for P > 1, per-orbit transfer traces, variance in P), with K4/K5 as corollaries and K1 as (partially pre-occupied) foundation. But the finite-field core of K1 — and possibly counting territory adjacent to K2/K4 — cannot be cleared while L1 is unread. A novelty claim built today would rest on an incomplete search against the single most relevant finite-field predecessor, written by (among others) this project's own advisor's group — the worst possible source to leave unread.

## 4. Novelty statements (Step 4)

### 4.1 Safest defensible statement (internal/planning use; adaptable for a manuscript only after §4.3)

> "In the sources we have been able to inspect in full or at abstract level (listed in P1_09_OVERLAP_MATRIX.md), we have not found a joint enumeration of labeled repeated-root constacyclic codes over product affine algebras by code dimension and Galois hull dimension, nor the per-orbit transfer-matrix trace formula and the associated mean and variance formulas in P proved here. Hull-dimension formulas for individual codes (including repeated-root cases under additional hypotheses) and counts of attainable dimension values are known (repo L4; repo L8 for multi-twisted codes), as are average hull dimensions over finite fields (repo L2) and small-hull-dimension criteria (repo L3). A finite-field hull-dimension formula and a count of codes by hull dimension are announced

in the abstract of Debnath–Prakash–Islam (2023) (repo L1), whose full text we have not been able to access lawfully; we make no priority claim relative to that paper."

### 4.2 Claims that must NOT be made yet

1. "First" hull-dimension formula/generator for repeated-root (or general) constacyclic codes — contradicted by L4 Thm. 7/8 and L8, and unverifiable vs L1.
2. "First" count of constacyclic codes by (Galois) hull dimension — L1's abstract announces such a count; scope unread.
3. Any priority claim over repo L1 on the finite-field specialization of T3/T4/C3/C4.
4. "First" average/mean hull dimension (L2), "first" small-hull-dimension results incl. p | n (L3), "first" repeated-root hull treatment (L3/L4/L9–L10 scope records).
5. "No prior work exists" / "novel" unqualified — the search is non-exhaustive (MathSciNet/Scopus/WoS unsearched; L1 unread).
6. Publisher-version claims for L4/L8 theorem numbers (only arXiv numbering verified; L8 publisher numbering demonstrably differs).

### 4.3 Specific missing source/comparison needed to strengthen the claim

One item: **a lawful full text of repo L1** (author-supplied copy — note the third author is this project's advisor, so an author/institutional channel is realistic — or library/ILL), followed by a theorem-by-theorem extraction answering L1-Q1…Q6 (§2) against T3/T4/C3/C4/C5 and the K2 joint-vs-marginal question. Secondary strengthening: publisher-version spot-verification of L4/L8 main theorems; a MathSciNet/Scopus pass if access becomes available.

### 4.4 Journal-level novelty-risk assessment

- **If submitted today** (e.g. to Cryptography and Communications∗, Advances in Mathematics of Communications∗, or ∗Finite Fields and Their Applications∗) with the §4.1 statement and honest positioning: risk is **medium** — the enumerator/transfer/moment package is a real contribution over the inspected record, but a referee who knows L1 (plausible at exactly these journals; L1 sits in Cryptogr. Commun.) could force major reframing if L1's count turns out to be joint or to cover p | n.
- **If submitted today with any "first" wording:** risk **high** — L4/L8/L3 already falsify broad versions, and L1 is a standing unknown.
- **After L1 is read:** risk drops to **low–medium** in the best case (L1 simple-root and/or marginal-only → our repeated-root affine joint package is cleanly positioned) and is **manageable** in the worst case (positioning shifts to the affine-product assembly, transfer traces, and P-moments, with L1 cited for the finite-field core). No acceptance probability is guaranteed or implied; this is a risk assessment, not a prediction.

## 5. Decision

- **Gate G1: DEFER (cannot close).** The blocking item is singular and identified (L1 full text). Closure criteria for a future vote: L1-Q1…Q6 answered from a lawful full text; overlap matrix updated; §4.1 re-checked word by word.
- **Overall: CONDITIONAL GO** — the proved, corrected, frozen mathematical core (T1–T6, C1–C6; blueprint SHA-256 ec2fd8d5…, commit 1c67fda) is unaffected by this audit; what is conditioned is any novelty claim and the manuscript phase that depends on Gate G1.
- **NO-GO for novelty/priority claims** until Gate G1 closes.

## 6. What was run / changed in this command

- Fresh lawful fetches: OpenAlex DOI record; Semantic Scholar Graph API record; web searches (L1 manuscript hunt — negative; targeted joint-enumerator/transfer hunt — no counterpart). No

paywall bypass; no theorem inferred from an abstract.

• Files created: P1_09_GATE_G1_NOVELTY_AUDIT.md, P1_09_OVERLAP_MATRIX.md (+ PDFs for side viewing). Files updated (sequentially): 04_BLUEPRINT_GAP_REGISTER.md, ARENA_WORKLOG.md. **Not modified:** 02_THEOREM_BLUEPRINT.md, all proof files, all validation scripts/outputs.

• Existing validation scripts were also rerun unmodified as a sanity check (outputs redirected outside the repo artifacts): validate_p1_03.py exit 0 and byte-identical to its saved output; validate_p1_05.py and validate_p1_06.py both printed their ALL-PASSED final lines. No saved output file was touched.